Overview and Comparative Analysis of Gas Turbine Models for System Stability Studies

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***Abstract*—Gas turbines have become increasingly popular in the different power systems, due to their lower greenhouse emission as well as the higher efficiency, especially when connected in a combined cycle setup. With increasing installations of gas turbines scheduled in different countries, the dynamics of the gas turbines become increasingly more important. In order to study such dynamics, accurate models of gas turbines are needed. This paper presents a comparative analysis and an overview of various models of gas turbines published in different literature.**

***Index Terms*—Gas turbines, models, overview.**

I. Introduction

Amid growing concerns about greenhouse emissions, gas turbines have been touted as a viable option, due to their higher efficiency and the lower green house gas emissions compared to other energy sources and fast starting capability which enables them to be often used as peaking units that respond to peak demands. Many power producers and governments have increased or are increasing their reliance on gas turbine installations whether as a simple cycle configuration or as part of a combined cycle plant (CCP).

This paper presents a short overview of gas turbine and governor models used in various power system studies with the aim to describe and consolidate various models. A brief direct comparison of two most widely used models of the gas turbine (IEEE and Rowen) suitable for small and large disturbance stability studies is also presented. The main differences between the two models are highlighted and the possible simplification of the Rowen model is considered.

II. Basic Background on Gas Turbines

A gas turbine usually consists of a compressor, with a combustion chamber and a turbine operating under the Brayton cycle [1], [2]. Fig. 1 shows the basic layout of a generic gas turbine. The gas turbine is operating in a simple cycle without any heat recovery.

The ideal Brayton cycle is made up of four completely irreversible processes [1]: 1-2 Isentropic Compression; 2-3 Constant Pressure heat addition; 3-4 Isentropic Expansion, and 4-1 Constant Pressure heat rejection. Air is first compressed in an adiabatic process with constant entropy within the compressor (process 1–2), usually an axial compressor. Pressure of 13–20 times [3] that of atmospheric is achieved after the compression stage. Fuel, either liquid of gas is then mixed with the compressed air and burnt in the combustor (process 2–3). After which, the hot gasses is allowed to expand through the turbine (process 3–4). This gas expansion drives the blades of the turbine and consequently the shaft of the generator connected to it.

As a part of the original assumption, the gas turbine modeled is assumed to only operate in a simple cycle with no heat recovery. However, this may not be always the case as there are more and more installations of CCP in order to exploit their high efficiency [3]. With CCPs the airflow is controlled to maintain a high exhaust temperature, even when partly loaded. The airflow is adjusted via inlet guide vanes (IGV), which change geometry to adjust the airflow from the compressor.

The single shaft gas turbine as its name implies, has all the masses (the compressor, combustor and turbine) connected on a single shaft. This makes the overall inertia of the gas turbine larger as compared with the latter aero-derivative type gas turbine. In addition, the aero derivative type gas turbine has the gas generator (compressor and compressor turbine) and the power turbine mechanically separated. As the compressor is on a different rotating shaft, different speed setting for the compressor can be achieved from those of the power turbine. This variable speed setting allows the gas turbine to achieve higher efficiencies at part load as compared to the single shaft gas turbine shoes efficiency reduces if operated away from the nominal point.

III. Existing Models

Many models representing the gas turbine have been developed over the years. A brief discussion of the existing models is given below.

A. Physical Models

Physical models derive the model directly from dynamic physical thermodynamic properties and laws. They involve utilizing laws governing thermodynamic behavior in the Brayton cycle [1], [2] along with some simplifying assumption to obtain the differential equations representing the dynamic gas turbine behavior. These laws include conservation of mass, conservation of power and conservation of energy [5]–[9].

Below is an example of the differential equations adopted from [7]. It shows how the differential equations are obtained using the physical thermodynamic laws and behaviour of the gas turbine. Equation (1) is the conservation balance of total mass

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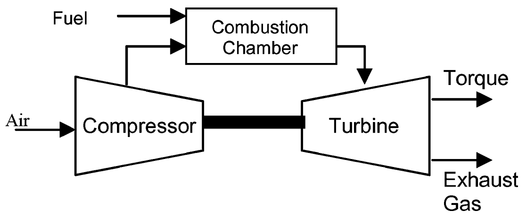

Fig. 1. Open cycle gas turbine engine.

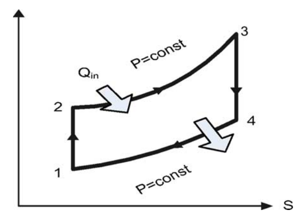

Fig. 2. T-S diagram.

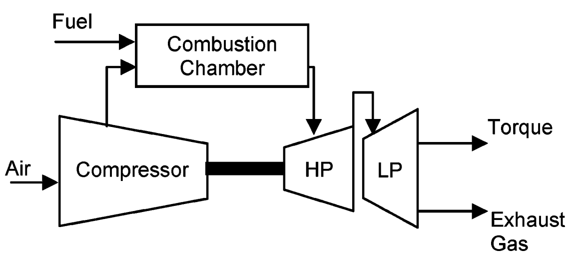

Fig. 3. Aero derivative/twin-shaft gas power turbine.

while (2) refers to the conservation balance of internal energy in the gas turbine

$$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out} \tag{1}$$

$$\frac{dU^*}{dt} = \dot{m_{in}} i^*_{in} - \dot{m_{out}} i^*_{out} + Q + W \tag{2}$$

where m represents the mass, U represents the internal energy, i refers to the specific enthalpy, Q is the heat input and W is the work done.

Over the years, many different types of gas turbine have been developed for different applications. For power generation, the gas turbines are essentially split into two distinct types; heavy duty industrial gas turbine or the single shaft gas turbine and the aero-derivative or twin/multiple shaft gas turbine [2]–[4]. The Fig. 3 shows the aero derivative gas turbine. The key cycle that is essential to the operation of a gas turbine is the Brayton cycle. To explain the Brayton cycle, the classic Temperature versus Entropy diagram is shown in Fig. 2.

Other authors (especially those having a mechanical engineering background) [5], [8]–[10] also utilize physical laws as well as thermodynamic laws in order to derive the equations representing gas turbine dynamics. They model different components of the gas turbine such as ducting, compressors, combustors and air blades [5], [8]–[10].

B. Rowen's Model

This model appeared in Rowen's paper [11]. It entails a simplified mathematical model for heavy duty gas turbines. The

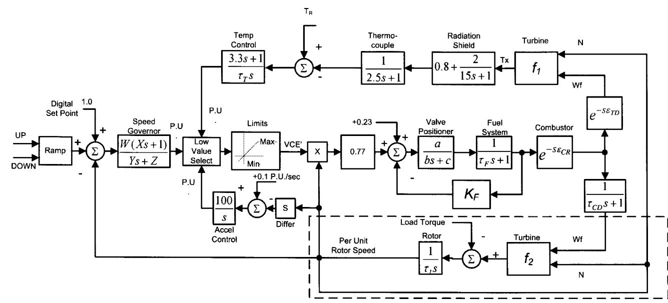

Fig. 4. Rowen's model (reproduced from [11]).

following assumptions were made: i) it is a heavy duty gas turbine operated in a simple cycle with no heat recovery; ii) fairly constant speed is maintained between 95%–107% of the rated speed; iii) it operates at an ambient temperature of 15°C and at an ambient pressure of 101.325 kpa. Since then, the model has been utilized to investigate the impacts of governor on system operation [12]–[14]. It was derived from, and validated against the actual operation [12] data and found to be adequate for a real life implementation [12], [13]. It is shown in Fig. 4 in a simplified block diagram format. The single shaft gas turbine along with the control and fuel system is represented. Control system of the gas turbine has three control loops; the speed control, temperature control and acceleration control. These three control functions are all inputs into a minimum value selector (represented by the low value select block). Output of the low value select represents the least fuel control actions among the three control actions. The speed control loop corresponds directly to the governor and can be operated either in the standard droop configuration or in isochronous mode. The temperature control loop represents the limitation of the gas turbine output due to temperature. Exhaust temperature is measured using a series of thermocouples incorporating radiation shields as shown in the model. An acceleration control loop, in order to prevent the over-speeding of the generator in the event of a sudden loss of load, is also implemented in the model and represented by the third input into the low value select.

Centeno in [15] provides a more detailed explanation and description of the functions of the different control loops as well as the modeling of the different control functions. Dynamics of the turbine in the Rowen's model is essentially made of the function blocks $f1$ and $f2$, the delays associated with transport of the exhaust gas and the combustion process as well as the time lag block with a time constant of $\mathrm{T_{CD}}$. Function block $f1$ (a function of fuel flow and rotor speed) calculates the exhaust temperature of the turbine. Block $f2$ calculates the turbine torque output of the gas turbine and again is a function of the fuel flow and rotor speed. The functions $f1$ and $f2$ are reproduced below from [11] where additional details can be found as well

$$f1 = TR - af1 * (1 - wf1) - bf1 * (speed) \tag{3}$$

$$f2 = af2 + bf2 * (wf1) - cf2 * (speed) \tag{4}$$

where $af1$, $af2$, $bf1$, $bf2$, $cf2$ represent coefficients and constants in the equations, while TR refers to the rated exhaust temperature, $speed$ refers to the speed deviation of the rotor and $wf1$ refers to the fuel flow.

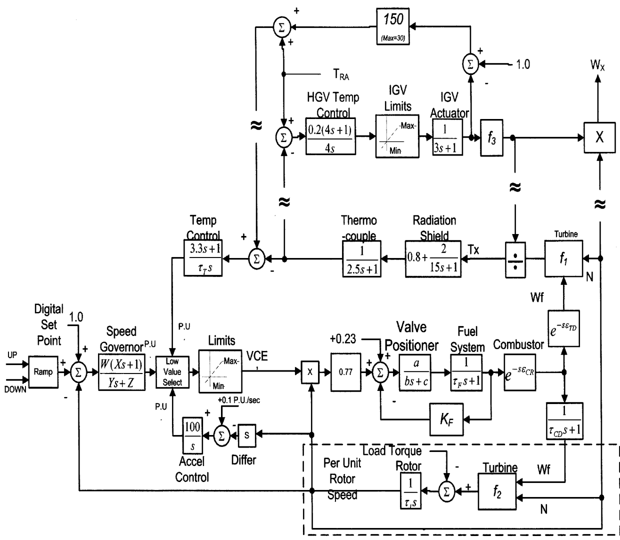

Fig. 5. Rowen's model with IGV (from [16]).

Rowen extended the original model (shown above) in [16], to include IGVs and their effect on the gas turbine dynamics, especially the exhaust temperature. Though the IGVs are included, the control loops for the speed and acceleration control remain essentially the same. The IGV controls can be seen in Fig. 5. Function f_1, which calculates the exhaust temperature, is now augmented to include the impacts of the changing airflow as well as ambient temperature. A new function f_3 is included in the model to calculate the exhaust flow calculation. Similarly to the first model [11], this new model has also been used to study the governor and the gas turbine operation. Since it enabled a more accurate modeling of a gas turbine operation installed as part of a CCP, many researchers have utilized it for studies involving CCPs [17], [18].

The augmented function f_1' is given in (5) at the bottom of the page. It can be clearly seen that it incorporates the effect of the IGV, fuel flow, ambient temperature and rotor speed, where Tr refers to the rated exhaust temperature, Ta refers to the ambient temperature, Wf represents the fuel flow and N refers to the rotor speed.

New function, f_3, given by (6) calculates the flow of exhaust gases from the gas turbine (necessary for the subsequent heat recovery stages of the CCP)

$$f_3 = N\frac{519}{T_a + 460}(Ligv)^{0.257} \tag{6}$$

where Ta refers to the ambient temperature, $Ligv$ represents the output of the IGV and N refers to the rotor speed.

Further simplifications of the Rowen's model can be made in order to suit the different operating conditions of different gas

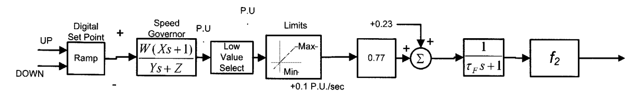

Fig. 6. Simplified Rowen model.

turbines. According to [16], for a simple operation of the gas turbine without any heat recovery, the IGVs would only be in operation during the start up and the shut down of the gas turbine. However, the IGVs are also modulated at part load operation when the gas turbine is used as a part of a combined cycle plant. During normal operation of the gas turbine, the IGV would usually not be in operation, hence, the portion pertaining to the IGV (shown in Fig. 5, above the $\approx$ symbol) can be omitted. With the IGV section omitted, the model would be reduced to the original one shown in [11]. The acceleration control loop can also be omitted if frequency variations are not greater than 1% (as the machine is operating in a relatively "stiff" system [11] where over speeding is unlikely to happen unless a load-loss situation arises) as well as the temperature control loop when the control of the gas turbine is dominated by the governor (diminishing the action of the temperature control signal). With only one signal remaining, the low value select can also be omitted.

Simplifications mentioned above are some of those that are possible with the Rowen's model. Other simplifications are also possible, which could make the reduced model even more simplistic [11]. An example of the simplification is shown in Fig. 6 where the acceleration and the temperature control loop are neglected. (*Note*: It should be noted that the torque equation, f_2, is only accurate to within 5% at part load and significantly more accurate at rated load. The exhaust temperature equation, f_1, is even less accurate at part load. However as the temperature is inactive during the part load operation, the inaccuracy of f_1 is considered to be negligible [11].) This particular simplification is applicable only when the gas turbine is operating at part load and far away from the peak output of the gas turbine.

Finally, since the papers [11], [16] are based on specific models of GE gas turbines, namely the 5001-9001 series, the various constants featuring in $(f_1 - f_3)$ need to be derived separately for other types of turbines. Liu demonstrates in [19] how different coefficients can be calculated based on the design characteristics of the different gas turbines.

C. IEEE Model

Findings from the IEEE task force (TF) on modeling of governors [20] formed the basis of the IEEE model. The same IEEE TF also published a series of papers on prime movers modeling [21], [22]. Essentially, the IEEE model is split into two parts: one pertaining to the controls of the gas turbine (the temperature control loop, the air flow control loop and the fuel flow control loop) and the other representing the thermodynamic properties of the turbine. The gas turbine modeled developed applies to

$$f_1' = \frac{Tr - 815(N^2 - 4.21N + 4.42)0.82(1 - W_f) + 1300(1 - N) + 3.5(MaxIGV - IGV)}{1 + (1 + 0.0027(59 - T_a))} \tag{5}$$

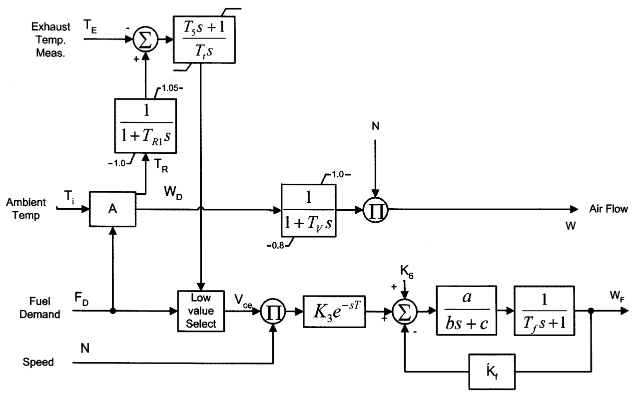

Fig. 7. Controls for IEEE model.

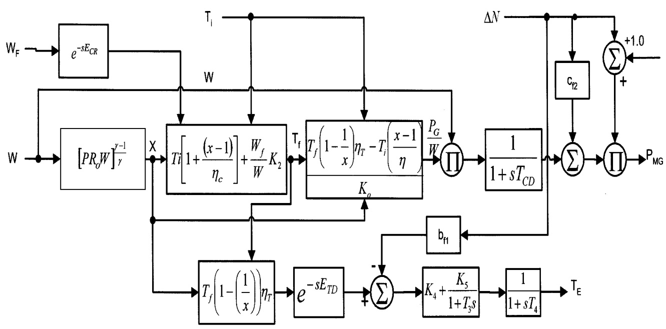

Fig. 8. Thermodynamic equations for IEEE model.

a constant speed simple cycle gas turbine with variable guide vanes in order to maintain a constant firing temperature for low greenhouse gases emission.

Comparison of the IEEE model to that presented in Rowen's first paper [11] reveals that the main difference is the control action necessary to maintain a high firing temperature (turbine inlet temperature). A beneficial by product of the high firing temperature is low NOx gasses emission level. (This action of the IGVs is included in the later Rowen's model [16].) The IEEE model assumed a fixed compressor ratio, which is only valid for a relatively constant rotor speed. Fig. 7 shows the corresponding control scheme of the gas turbine which is similar to that shown in the Rowen's model previously. However, as the modulating actions of the guide vanes are modeled [20] an additional airflow control loop is included. (This is similar to the inclusion of the IGV controls in Rowen's model for mechanical drive [16].) Additionally, the control block A, which essentially schedules the air flow, is a nonlinear function of (8) and (9). The calculation required by block A have to be solved via solvers such as the Newton Raphson method due to the nonlinear nature of (8) and (9).

Fig. 8 below shows the block diagram representing the necessary calculations in order to derive the mechanical power output of the gas turbine. The connections in the block diagram are essentially based on the isentropic efficiencies equations [2] for the compressor and the power balance equation. The Ecr and Etd parameters (in Fig. 8) refer to the time constants related to the time delays. The following equations are stated below for the completeness of the discussion:

$$T_R = T_f\left[1-\left(1-\frac{1}{x}\right)\eta_T\right] \quad (7)$$

$$x = [PR_0 W]^{\frac{\gamma-1}{\gamma}} \quad (8)$$

$$W = \frac{P_G K_0}{T_f\left(1-\frac{1}{x}\right)\eta_T - T_i\frac{(x-1)}{\eta_C}} \quad (9)$$

$$T_f = T_D + \frac{W_f}{W}K_2 = T_i\left(1+\frac{x-1}{\eta_C}\right) + \frac{W_f}{W}K_2 \quad (10)$$

where

T_R	reference exhaust temperature;
x	cycle pressure ratio parameter;
PR_0	design cycle pressure ratio;
γ	ratio of specific heat capacities;
W	air flow;
η_T and η_C	represent the turbine and compressor efficiency, respectively;
T_f	firing temperature;
T_i	ambient temperature;
T_D	compressor discharge temperature;
W_f	fuel flow.

Based on the isentropic efficiencies of the compressor and the turbines, (7) and (10) can be derived accordingly. The ideal adiabatic process is isentropic, however, in reality this is not exactly the case. This results in inefficiencies in the adiabatic processes, i.e., turbine and compressor isentropic efficiencies ($\eta_{\rm T}$ and $\eta_{\rm C}$) respectively. A brief derivation based on the compressor efficiency is shown below while more details can be found in [1], [2]. Bearing in mind that the temperature change for the ideal compressor cycle is $T_{02} - T_{01}$ (Temperature of cycle 1–2 in Fig. 2) the real temperature change would be $T'_{02} - T_{01}$. Compressor isentropic efficiency is defined as

$$\eta_C = \frac{C_p \Delta T'_0}{C_p \Delta T_0} = \frac{T'_{02} - T_{01}}{T_{02} - T_{01}} \quad (11)$$

where Cp refers to the mean heat capacity of the gas and T'_o refers to the change in temperature in reality while T_o refers to the temperature change in the ideal process.

Pressure ratio of the cycle is linked to temperature [1], therefore (11) can be further developed into

$$T_{02} - T_{01} = \frac{1}{\eta_C}\left(T'_{02} - T_{01}\right) = \frac{T_{01}}{\eta_C}\left(\frac{T'_{02}}{T_{01}} - 1\right). \quad (12)$$

Neglecting the pressure loss in the combustor, P_{02} is equal to P_{03}. Therefore the ratio of P_{02}/P_{01} is equivalent to the cycle pressure ratio, x; turbine inlet pressure divided by ambient pressure. Hence T_{02}, which is T_D (compressor discharge temperature), can be calculated by (14). Considering that T_{01} as the ambient temperature (14) can be seen to be equivalent to the T_D section in (10)

$$T_{02} - T_{01} = \frac{T_{01}}{\eta_C}\left(\left(\frac{P_{02}}{P_{01}}\right)^{\left(\frac{\gamma-1}{\gamma}\right)} - 1\right) \tag{13}$$

$$T_{02} = T_{01} + \frac{T_{01}}{\eta_C}(x-1). \tag{14}$$

Using turbine isentropic efficiency instead, (7) can be similarly worked out as shown above. A detailed derivation of both equations is shown in [2]. It should be noted that the temperatures in the IEEE model are worked out in absolute temperature, K°.

The IEEE model has also been further developed in [23] to take into account effects such as the use of a part of the overall air flow in order to cool the turbine blades. The enhanced model [23] was found to be able to simulate the dynamic behavior of the gas turbine with the required accuracy.

D. Aero-Derivative Model

The previous models focused on heavy duty or single shaft gas turbines where the rotational masses are connected on a single shaft. As there are many aero derivative gas turbines connected to the network (especially for smaller machine ratings), a model for an aero-derivative gas turbine was also developed. These gas turbines are essentially derived from jet engines and often utilized to exploit their better efficiencies at part load operation compared to the heavy duty gas turbine. As in this case the compressor is connected on a different shaft as compared to the power turbine, it is generally able to rotate at different speeds and achieve better compressor ratio and hence better overall performance.

Fig. 9 shows the overall block diagram for the twin shaft gas turbine adopted from [24]. The gas turbine is split into the engine (connected to the compressor) and the turbine (the low pressure turbine connected the the generator shaft). In terms of the format of the block diagram it is similar as that of the Rowen's model. However, instead of a single speed signal going into the low value selector, there are now two speed signals. One is the speed of the engine (which essentially determines the speed of the compressor) and the other is the speed of the turbine (which corresponds to the speed of the low power turbine or the Generator). Also, similar to the Rowen's model, the turbine characteristics indicated by functions $f_1 - f_4$ in the twin shaft model (Fig. 9), can be obtained from the operation characteristics of the actual turbine itself. These functions are derived based on the operating curves as shown in [24]. Operating curves such as the electrical power versus fuel flow, or the exhaust temperature versus fuel flow, can be used to obtain the different functions and the various model parameters stated above. The ultimate model parameters are then obtained through a trial and error process in order to fine tune the parameters till the responses are matched to the actual gas turbine response.

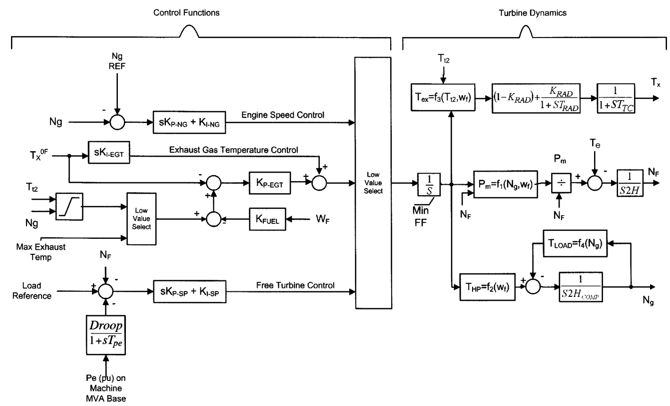

Fig. 9. Twin shaft turbine (reproduced from [24]).

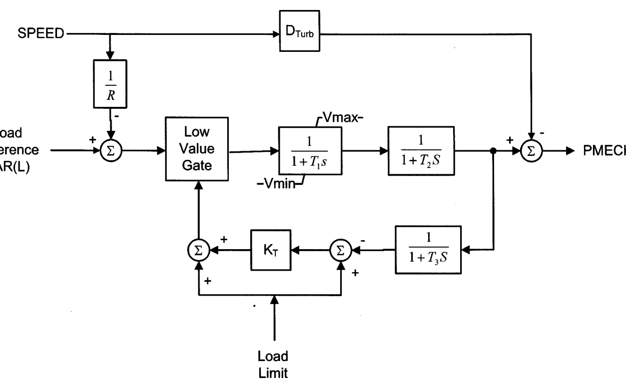

Fig. 10. GAST governor model.

E. GAST Model

Fig. 10 shows the GAST model which was one of the most commonly used dynamic models for the governor [14]. This was partly due to the simplicity of the model and partly due to the fact that it was once WECC compliant. Even though it was one of the most widely used models it has now been found to be deficient in certain areas [14]. It is unable to model the gas turbine operation accurately when the temperature control loop becomes active, and to replicate system oscillations around the final settling frequency.

The GAST Model is no longer WECC compliant and it has been superseded by other, more accurate models such as the GGOV1 model discussed in the next section.

F. WECC/GGOV1 Model

Over the years, numerous trips of large generating plants in the Western Electricity Coordinating Council (WECC) have been observed. This has been attributed to the inaccurate modeling of many of the thermal unit governors (including Gas Turbines) in the WECC. Since early 2001, WECC has proposed new criteria for Frequency Responsive Reserves (FRR) making the need to develop a more accurate governor model for dynamic simulations imperative [25]–[30]. A model shown in Fig. 11 was developed using measured responses and data collected from two trip tests performed on the 18th May 2001. In separate tests, 750 MW and 1250 MW of generation were tripped in the Southwest and Northwest respectively, while the AGC was turned off. The tests had revealed that

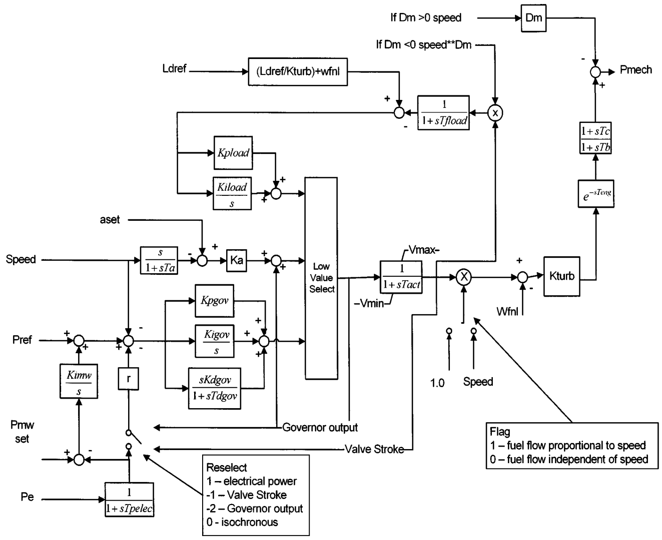

Fig. 11. GGOV1 model (reproduced from [30]).

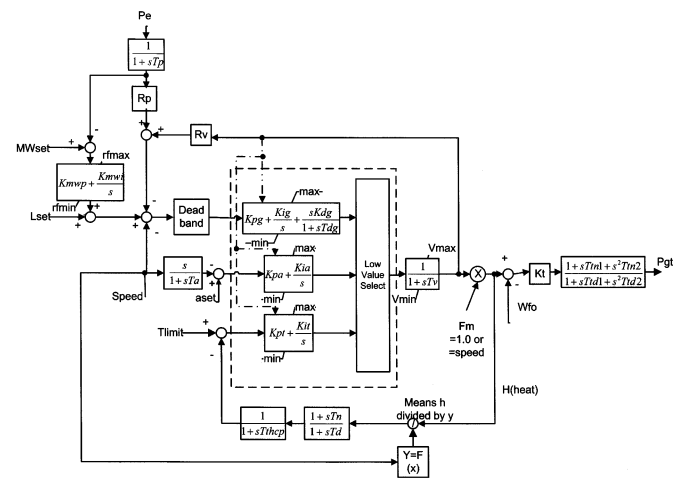

Fig. 12. CIGRE model (reproduced from [3]).

only 40% of the expected governors responded (based on the simulations). The principal reason for the large discrepancy between the expected and measured governor responses is that base loaded and load limited turbine units were not modeled properly. These units were found to be predominantly thermal units including gas turbines.

Using a block diagram format, thermal units were modeled with separate elements such as the governor element, supervising element and the load management element. It should be noted that the GGOV1 model is a general model for all thermal units and the developed model can be utilized for representing gas turbines with suitable parameters in the various control blocks. Governor element essentially pertains to the basic governor and is a typical proportional-integral-derivative configuration. The droop can be implemented via a feedback signal of valve position or electrical power. The supervising element represents a load limit imposed by the operation of the power plant and in case of a gas turbine; the supervising limit represents the exhaust temperature limit. $\mathrm{L_{dref}}$ in the model represent this load limit and is given in terms of turbine power instead of exhaust temperature directly. This limit is imposed in the gas turbine model with a curve that relates exhaust temperature to several other engine variables [25]–[30]. The load management element will regulate the turbine power to the setpoint $\mathrm{P_{mwset}}$, simulating effectively the adjustments necessary due to the AGC commands. Turbine dynamics is essentially considered to be in direct proportion with fuel flow (the relevant time constants are longer compared to the turbine time constants) and hence can be simply represented by a single lead-lag block. Many of the GAST models previously used in the system were replaced by the developed GGOV1 model which is explained in further detail in [25]–[30].

G. CIGRE Model

Recognizing the increasing importance of the gas turbines, a CIGRE Task Force for gas and steam turbines in combine-cycle power plants, has developed a model of combined cycle power plant as shown in Fig. 12 [3], [31]. Similar to the Rowen's model, there are three major control loops feeding into a low value select. These control loops represent the speed/load governor, the acceleration control loop and the temperature control loop. However there are certain differences in the control representation. There is an additional outer loop plant control represented by the MWset to maintain the unit's output at a pre-specified MW level [3]. The minimum fuel flow in this model is represented by the limit Vmin, i.e., differently from the Rowen's model. Turbine dynamics is modeled by the second-order block instead of calculating the torque function as in Rowen's model.

The exhaust temperature is not explicitly calculated, instead, the temperature control is provided via a signal calculated via a function F(x) as shown in the figure below. This function is obtained from curves which relate exhaust temperature to turbine variables such as rotor speed. By choosing appropriate values for different control parameter constants, any desired mode of governor action can be simulated. A set of example parameters for the gas turbine model can be found in [3]. The CIGRE model is similar to the GGOV1 model described in the previous section, however, it specifically models a gas turbine instead of a generic thermal unit (hence the explicit modeling of the temperature limit).

H. Frequency Dependent Model

Many of the models mentioned in the previous sections are not suitable for determining the frequency dependency of the gas turbine [32]. To be able to analyse incidents with abnormal system frequency behavior, the frequency dependence of the gas turbine model must be taken into account. This was the main aim of [32] and a model which is based on the physical principles is developed in order to clarify the effects that shaft speed and ambient temperature has on shaft speed. A brief explanation of the various effects that frequency and ambient condition have on power output is shown in [33]. Changes in frequency are equivalent to changes in shaft speed and would result in a change in airflow. This change then translates firstly into a change in the pressure ratio across the compressor and secondly into a change in fuel level (in order to maintain the given firing temperature). These changes will directly affect the maximum power output [33]. A similar relation is reported between the ambient temperature and the maximum power output, however changes in ambient temperature have a much more severe impact compared to

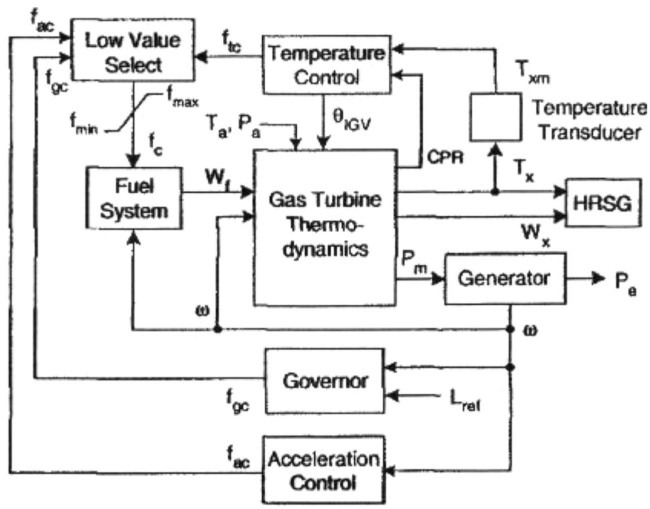

Fig. 13. Frequency dependent gas turbine (reproduced from [32]).

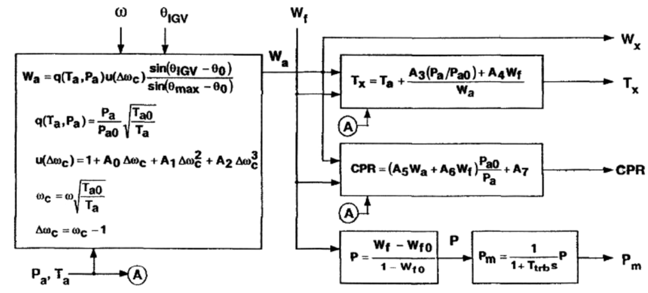

Fig. 14. Thermodynamic equations for frequency dependent gas turbine (reproduced from [32]).

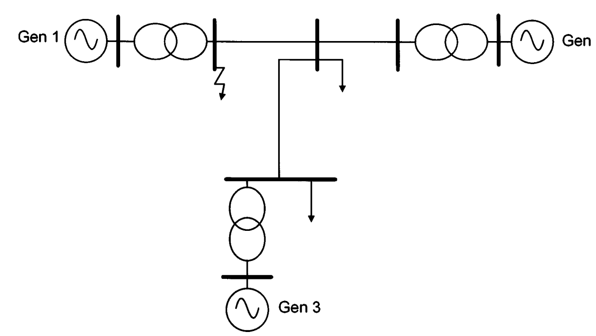

Fig. 15. Test system diagram.

changes in rotor speed. The paper [33] also discusses the characteristics of the axial compressor and the physical principles of the gas turbine.

The overall block diagram of the frequency dependent model is shown in Fig. 13. Again, the control scheme is similar to that found in the previous models, such as the Rowen's model. Fig. 14 shows the thermodynamic equations representing the dynamic behavior of the gas turbine. Unlike the Rowen's model, where only the output power and the exhaust temperature were necessary, this model calculates the compressor pressure ratio and the exhaust gas flow in addition to exhaust temperature and mechanical power output. Equations representing the impact of IGVs have also been incorporated into the model. The various parameters of the model (e.g., A_0, A_1 and A_2, etc.) are obtained directly from the test data of actual machines. Based on this model, a CCP model for investigating frequency excursions was developed and tested in [34]. Malaysia black out was quoted in the paper as an example of the abnormal frequency event, as well as the formation of electrical power islands with a power imbalance. This study found that the dependency of the output of the gas turbine on frequency and ambient temperature is significant and that both, the temperature control and the governor play critical role during such abnormal frequency operations. Interestingly enough, the frequency dependent model is based on similar equations to those used in the IEEE models. However, instead of a fixed compressor ratio with small deviations, the frequency dependent model assumed a generic form representing the dependence of the pressure ratio on frequency deviations as well as ambient temperature.

IV. Comparative Analysis of the Key Models

The previous section presented an overview of various gas turbine models currently available for system dynamic studies. Their different degree of complexity makes them suitable for different types of studies. The most complex ones, the physical models, are best suited for the analysis of the specific mechanical and thermodynamic behavior of individual gas turbine. However, they are far too complicated for large system studies.

Each of the models presented here, i.e., Rowen, IEEE, CIGRE and GGOV1, is suitable for system studies. The actual turbine control representation however, must be carefully verified to ensure that the selected model is adequate for the intended study. These models are still able to capture relevant dynamics pertaining to the power system even though they are not as detailed and complicated as the physical models.

The main aim of this particular study (as indicated in Section I) is to identify and critically assess suitable models for system stability studies, especially those for transient and small disturbance stability studies. The IEEE and the Rowen's model have been chosen as they are some of the most commonly utilized models for this type of system studies. They are subjected to further investigation and comparison in the sequel.

A. Performance of the IEEE and the Rowen's Model

In order to compare the performance of the IEEE and the Rowen's model, a simple 3 machine network is utilized. The test system is shown in Fig. 15. A three phase self-clearing fault is simulated at the high voltage end of the transformer terminal connected to generator 1. With the gas turbine model (the IEEE and Rowen's, alternatively) connected to generator 2, the output of the generator 2 as well as the control inputs to the gas turbine are plotted and compared. (*Note*: The acceleration control loop was not modeled in either of the models.)

Typical generic gas turbine parameters are used in the Rowen's model [11], [12]. Parameters of the control mechanisms for the two models are set to be identical. Controls for the both models are also almost identical except for the air flow control loop in the IEEE model. (In the simple cycle gas turbine model without IGV action, i.e., original Rowen's model [16], the air flow is essentially assumed to be fixed or with a very little variations. Hence, the air scheduling for low emission in the IEEE model is disabled and the air flow is set to be at the nominal level at all times, i.e., 1 p.u.) Since

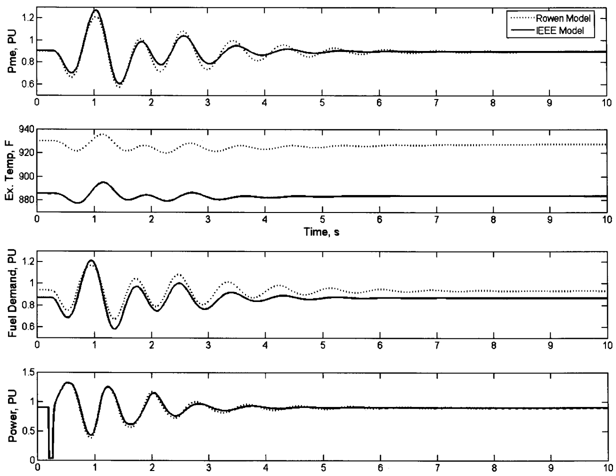

Fig. 16. System responses with typical Rowen model and IEEE model.

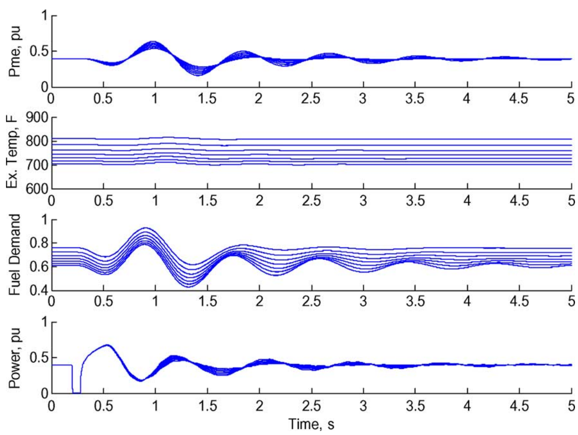

Fig. 17. Responses of system with variation in torque coefficients.

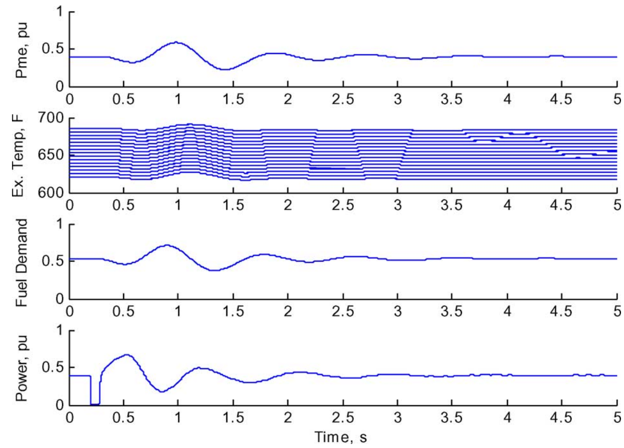

Fig. 18. Responses for variation in exhaust temperature coefficients.

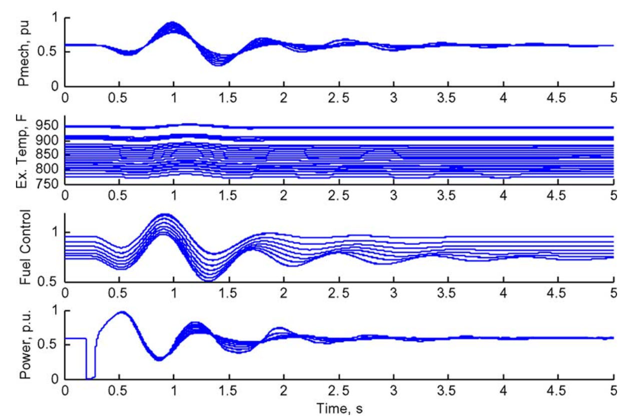

Fig. 19. System responses for variation in coefficients of the torque and exhaust temperature equation.

the controls are virtually the same, the eventual differences in model performance resulting from different model parameters can be fully attributed to respective gas turbine representations.

The obtained responses with two models are compared in Fig. 16. Different behavior of the two models is evident.

Even though the Rowen's model was originally designed for a specific genres of the GE gas turbines with corresponding parameters [11], those parameters have been utilized in the past to represent generic gas turbines in various studies [35], [36] including this one. The parameters of the IEEE model used in this study were derived using the data from a SIEMENS Gas Turbine brochure.

In the Rowen's model, the gas turbine dynamics is represented by two functions representing the torque and the exhaust temperature (see previous chapter), respectively. By varying the coefficients of those functions, the sensitivity of the model to these coefficients can be established. This is first done for the torque equation by changing the coefficient values by $\pm 10\%$. The results showing the variation in the power output of the gas turbine are presented in Fig. 17.

For the same range of variation of the coefficients in the exhaust temperature equation the rated exhaust temperature remains unchanged as shown in Fig. 18. The exhaust temperature curves can be seen to scale up and down according to the changes in the coefficients, however the control action and the power output from the gas turbine remains unchanged. This is due to the relatively low power output of the gas turbine (at 0.4 pu), where the temperature control does not activate. As the temperature control is not in operation, the output of the gas turbine is effectively determined by the governor only and hence the seemingly unaffected power output of the gas turbine.

The coefficients of both, the torque and the exhaust temperature equation, are then varied simultaneously. System responses are shown in Fig. 19. This resulted in larger variations in the gas turbine output compared to those shown in Figs. 17 and 18. For certain combinations of coefficient values, the temperature control loop was active causing the overall system response to be much more oscillatory than before. The model therefore is sensitive to variations in the coefficients of the gas turbine dynamic equations. This is particularly so for the torque equation. Changing the coefficients of the torque equation has an impact on both the power output and the exhaust temperature of the gas turbine. The coefficients of the exhaust temperature equation influence turbine responses only at higher loading levels.

Similar analysis was performed with the IEEE model. The parameters governing the gas turbine dynamics are varied. Those were x (Compressor Ratio), μ_{C} (Compressor Isentropic efficiency), μ_{T} (Turbine Isentropic efficiency) and K_2 (design combustor temperature rise). However, since the efficiencies are rel-

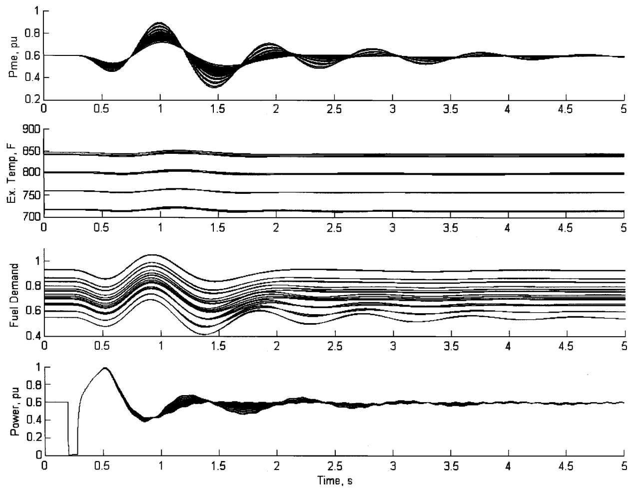

Fig. 20. System responses for variation in variables for the IEEE model.

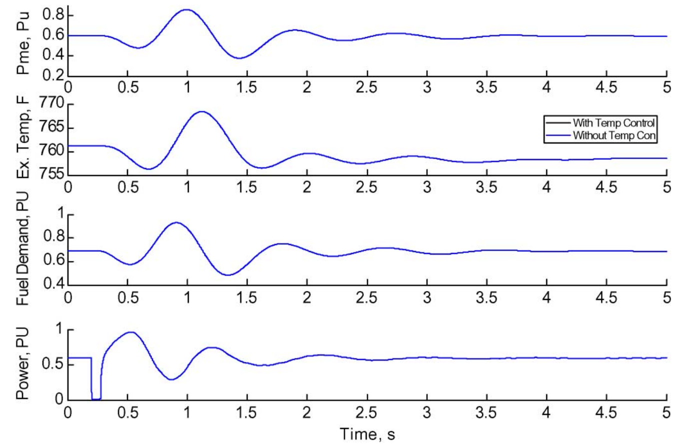

Fig. 21. Gas turbine responses with and without temperature control loop (0.6 Pme).

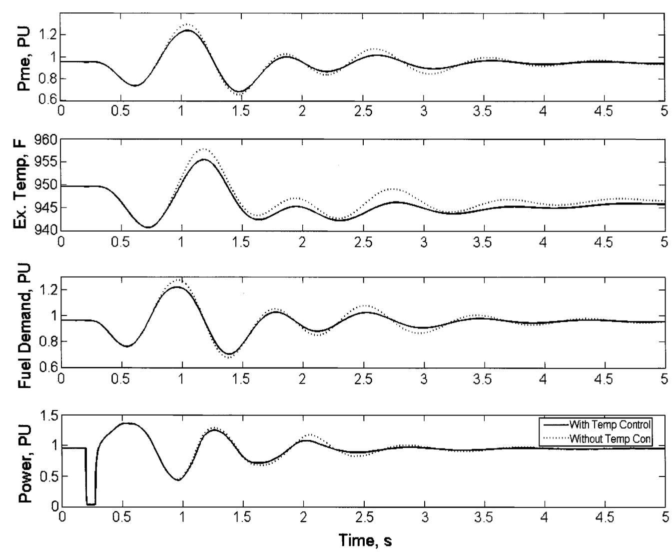

Fig. 22. Gas turbine responses with and without temperature control loop (0.95 Pme).

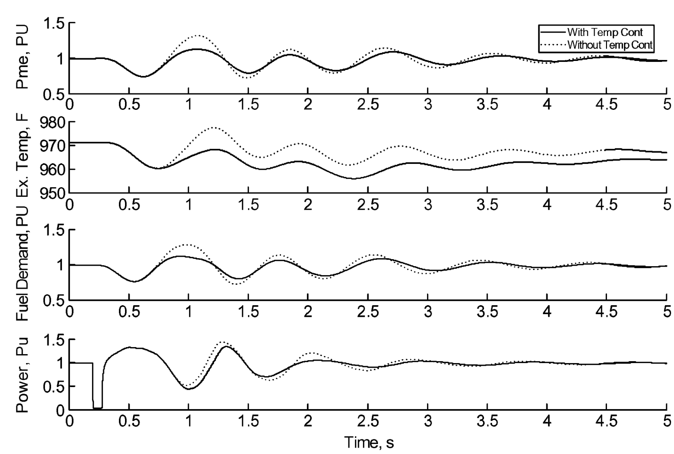

Fig. 23. Gas turbine responses with and without temperature control loop (0.99 Pme and 0.55 H).

atively fixed in real life, all of the above, except for the compressor and turbine efficiencies, are varied by only $\pm 10\%$.

Unlike the Rowen's model, the calculation of the gas turbine dynamics in this case does not involve functions $f1$ and $f2$; instead it is interconnected through the thermodynamic equations (see previous section). Hence, both, the exhaust temperature and the turbine torque are affected simultaneously if any of the above parameters (see beginning of the paragraph) is varied.

The results of simulations are shown in Fig. 20. It can be seen that the model is indeed sensitive (possibly even more than the Rowen's model) to variation in parameters. Variation of $\pm 10\%$ in system parameters causes larger change in the model behavior (0.18 variance in Pme) compared to that of the Rowen's model (0.15 p.u. variance in Pme).

It should be noted that the thermodynamic equations representing the gas turbine dynamics are linked, whereby a change in one variable would result in a corresponding change in the other. Hence, in order to maintain the nominal operation condition, the variables need to be balanced (variables have a direct corresponding correlation whereby changing one variable will require the other variable to be recalculated in order to maintain the nominal condition).

As mentioned previously, the Rowen's model can be simplified. After disabling the temperature control loop the simulations are repeated and the results compared in Fig. 21. The power output of the turbine was 0.6 p.u. It can be seen that there is no difference between the curves with and without temperature control. This is expected, as the operating point is far away from the rated exhaust temperature limit due to the low power output of the gas turbine. Hence, the temperature control loop is not active and makes the presence of the temperature control superfluous. Under such conditions therefore, the temperature control loop can be neglected without any loss in model accuracy. With a higher power output level (0.95 p.u.) though, the temperature control becomes active as the gas turbine is operating closer to the temperature limit. The comparison between the models with and without the temperature control loop in this case is shown in Fig. 22. Clearly, there is a difference between the responses with and without the temperature control loop.

The difference might not be as pronounced as that shown in Fig. 23 in case of small magnitude oscillations. However, if the system is subjected to a large disturbance the difference in responses with and without temperature control would be much more significant. This is illustrated in Fig. 23 where higher magnitude oscillations are obtained by simultaneously increasing the power output level from 0.95 to 0.99 and reducing the inertia constant of the machine from 0.6 to 0.55. In this case, the temperature control loop heavily dominates the gas turbine response and significantly influences the output of the gas turbine. (With higher loading levels, the gas turbine is much more sensitive to perturbations in speed.) Under these conditions, the modeling of the temperature control is essential as the accuracy of the model is seriously affected by the temperature control loop. It is therefore imperative to include the temperature control loop in the model.

B. Discussion of the Results

Clearly, there are differences between the IEEE and the Rowen's model. With different values for the coefficients in the gas turbine representation, the gas turbine behavior is extremely different. With the Rowen's model, the characteristics represented by the various functions are derived from operating curves which might not be easily available. However, the IEEE model can be derived if the exhaust temperature, ambient temperature and firing temperature for the nominal conditions, as well as the nominal compressor ratio are known. This information can be often found in commercial gas turbine brochures, which makes the derivation of the model for different makes of gas turbine easier. Though the derivation of the IEEE model might be easier, the equations are relatively complicated and could cause a problem in computation time especially for large systems. A compromise should be found therefore, between the two models. This issue will be addressed in the future research.

V. Conclusions

The first aim of this paper was to provide an overview of existing gas turbine models. To that end, different gas turbine models are identified, presented and discussed. The identified models are of different level of accuracy, suitable for different types of studies and have been utilized for different purposes in the past.

Among those, physical models are the most complex and the most accurate ones. They are suitable for detailed study of the dynamic behavior of the gas turbine. However, these models are too complex and unsuitable for the use in large power system studies.

For a more detailed analysis of power system and governor behavior, especially for equipment specific studies and large frequency excursions, models such as that in [32] should be utilized.

A brief comparative study of the IEEE and Rowen's model, reported in the paper, illustrated model dependencies on critical parameters and showed that the frequency and ambient temperature dependence of the gas turbine can significantly affect its operation under certain operating conditions. The frequency dependent model [32] should be used in particular, in the case of weak systems with large frequency variations.

Finally, the paper emphasizes that for modeling the aeroderivative gas turbines, often installed as a part of a combined cycle plant, the twin shaft model should be used. Obtaining the data to derive the functions required for the model, however, could be relatively difficult due to the requirement for separate compressor and high power turbine operation curves.

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